

Name:

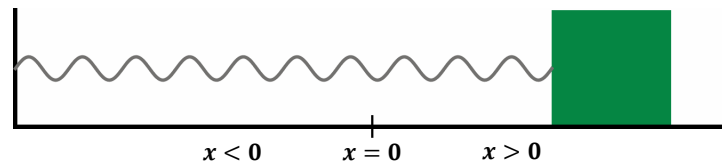
Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Examine Nullclines of 1st-order System of ODEs

1. Converting a linear 2nd-Order ODE into a Linear System of 1st-Order ODEs

This problem involves using Newton's Law of motion to formulate a set of rate equations that can describe, explain, and predict the motion of a mass on a spring.



The ODE that models the horizontal motion of the mass is governed by the 2nd-order linear homogeneous equation. Specifically with mass 1, friction coefficient 3, spring constant 2, and assuming no external forces, the ODE is given by

$$x'' + 3x' + 2x = 0$$

where x is the displacement of the mass at time t .

We can convert the 2nd-order ODE into a linear system of 1st-order ODEs which we can analyze. The following tasks leads you to convert the given 2nd-order ODE into a system of 1st-order ODEs.

Task	Show work here
Set $x' = v$ (velocity) and write the term for v' (acceleration).	
Substitute the established terms into the ODE.	
Isolate v' .	
Write x' and v' together by arranging them side by side as a system of equations, so it's clear how they relate to each other.	

The result is called a *linear system of first-order ODEs* because each equation involves only 1st derivatives of the dependent variables, and those variables can appear in any of the equations, possibly influencing one another.

2. Solving for the Nullclines of 1st-order System of ODEs

The *nullcline* was defined as an *isocline* where the slope (1st-derivative) is zero. In other words, it is a line that describes the critical points of an ODE.

Consider the same linear 2nd-order homogeneous ODE in Problem (1),

$$x'' + 3x' + 2x = 0$$

where it models the motion of a horizontal spring-mass system.

Setting $x' = v$ and $x'' = v'$ transforms the linear 2nd-order homogeneous ODE into a linear system of 1st-order ODEs,

$$\begin{aligned} x' &= v \\ v' &= -2x - 3v \end{aligned}$$

- | | |
|--|---|
| <p>a. The <i>x-nullcline</i> is when the derivative of the 1st equation is zero. Set $x' = 0$. Then, solve for v. Describe this line on an x-v axis.</p> | <p>c. Each nullcline describes the behavior of the system at its critical points.</p> <p>i. <i>x</i>-nullcline: Use your results in Part (a) to interpret what this nullcline means in context.</p> |
| <p>b. The <i>v-nullcline</i> is when the derivative of the 2nd equation is zero. Set $v' = 0$. Then solve for x. Describe this line on an x-v axis.</p> | <p>ii. <i>v</i>-nullcline: Use your results in Part (b) to interpret what this nullcline means in context.</p> |
| | <p>d. If the <i>x</i>-nullcline and <i>v</i>-nullcline intersect, what does this point of intersection represent in the context of a horizontal spring-mass system?</p> |